

Recognize and decode QR codes from images*

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Abstract — As digital technologies gain space in our daily lives, the use and recognition of QR codes becomes more and more important. This technical article examines the process of recognizing and decoding QR codes from images, with particular regard to innovative image processing methods. At the beginning, we review the history, structure and areas of application of QR codes. After that, we present the basics of image processing and the various techniques used to recognize QR codes, including image preprocessing, code location, segmentation, perspective correction, and decoding.

This technical article summarizes and interprets research results and developments in the field of QR code recognition and is a useful resource for researchers, developers, and industry professionals interested in QR code technologies.

I. INTRODUCTION

The history of QR codes dates back to the 1960s, when in Japan, due to the proliferation of supermarkets, people demanded the development of a technology that would facilitate the work of cashiers. With the cash registers used at the time, the prices of the products had to be entered manually, which was burdensome for the cashiers. The invention of barcodes alleviated this problem, but it soon became apparent that barcodes could store limited information. The development team of DENSO WAVE, in response to the needs of users, began to develop a new two-dimensional code that can store more information than traditional barcodes [1].

The main goal of Masahiro Hara, the specialist responsible for the development of the QR code, and his team was to develop a code that can be read quickly and can store a large amount of information. One of the biggest challenges during the development of the QR code was how to read the code as quickly as possible. Hara realized that the solution to this problem lay in the use of position sensing patterns. With the help of samples, it became possible to read the code at high speed. A year and a half after the start of development, the QR code was created, which was able to store 7,000 digits as well as Kanji characters and can be read more than 10 times faster than other codes.

In 1994, DENSO WAVE announced the release of the QR code. QR stands for "quick response", which emphasizes the quick reading of the code. Originally used in electronic Kanban systems in the automotive industry, the QR code has become essential for many companies for transparency and tracking. DENSO WAVE has decided to make the QR code specification publicly available so anyone can use it freely.

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In 2002, the use of QR codes became widespread among the Japanese public when mobile phones began to offer a QR code reading function. People could use QR codes to access websites or get coupons, which made using the technology extremely convenient. Today, QR codes have become indispensable tools in business and people's daily lives, and are used in a variety of ways, such as issuing business cards and electronic tickets, as well as in airline ticketing systems.

QR code is an open code that can be used by anyone, and it has spread not only in Japan, but all over the world. Rules for the use of the code were formulated and the code was standardized. In 1997, AIM (Automatic Identification Manufacturers) declared it a standard for the industry, in 1999 it was also accepted by the Japanese Industrial Standards, and in 2000 it was made an international standard by ISO (International Organization for Standardization) [2]. Currently, the use of QR code is so widespread that it is not an exaggeration to say that it is used all over the world [3,4].

II. AREAS OF USE OF QR CODE

QR codes have become indispensable in many areas of life, as they allow us to quickly and easily access digital information.

In advertising and marketing, QR codes are often used to direct consumers to a company's website or even to a product's information page. After scanning the codes, users can immediately access coupons and promotions offered by the company [5].

QR codes also play a big role in payment transactions. Mobile payment systems such as Apple Pay and Google Pay use them to simplify transactions [6].

In the field of tickets and entrance tickets, QR codes often serve an identification function. Flight tickets and concert tickets often contain QR codes that make entry easier and faster [7].

In retail, QR codes allow customers to quickly access detailed information about products, so they can make more informed purchasing decisions. In education, QR codes are often used to provide quick access to online learning materials such as notes, videos or quizzes [8].

In healthcare, QR codes can be useful for tracking patient data, identifying medications, and digitally displaying vaccination certificates and COVID-19 test results [9,10].

In personal use, QR codes are often used to share Wi-Fi network data or even to digitize business cards. In logistics and transportation, companies can use QR codes to track packages and simplify warehouse management [11,12], but legal regulations must also be taken into account [13].

Tourist attractions and museums often use QR codes to display additional information and multimedia content available to visitors. And restaurants can use QR codes to display their digital menus, which reduces physical contact and the use of paper [14,15].

In the automotive industry, car manufacturers can use QR codes to store car data so that service and maintenance can be faster and more efficient [16].

QR codes are useful in many other areas thanks to their flexibility and versatility. Codes allow users to quickly access digital information, whether it's product information, tickets, payment transactions, educational materials, health data, personal data, logistics, tourism information, restaurant menus or automotive data.

III. BACKGROUND AND THEORETICAL FOUNDATIONS

In the 1990s, developers working for various companies were trying to pack as much information as possible into their 2D codes. Traditional barcodes encoded information in one dimension, while 2D codes could encode information in two dimensions, i.e. horizontally and vertically. The biggest challenge for Hara and his team was making their code readable as quickly as possible. Hara realized that adding positional information to the code could solve the problem. This is how the position sensor pattern was created, which consists of square signals. This pattern allowed high-speed reading. The square shape was chosen because it is the least likely to be confused with other designs on various business forms. They developed a unique ratio between black and white areas (1:1:3:1:1) that could be detected regardless of the reading angle. After a year and a half of development, the QR code was created [3].

A. Structure and operation of QR codes

QR codes are square stickers consisting of dark cells (dots) on a light background. It also allows for reversed reflectances, i.e., light cells on dark background. These cells represent the data in binary form. QR codes have several different versions that differ in size and the amount of data stored. The smallest QR code consists of 21x21 cells, while the largest consists of 177x177 cells. The more cells a QR code consists of, the more data it can store.

The structure of QR codes consists of the following elements:

Positional patterns: These are large squares in the upper left, upper right and lower left corners of the code. They help the reader find the QR code and adjust the reading accordingly.

Timing Patterns: These form a row of cells that connect the patterns defining the upper left and lower left positions, and the patterns defining the upper left and upper right positions. Timing patterns help the reader determine how the cells are arranged within the QR code.

Format information: The format information cells store the level of data error correction and the masking pattern of the QR code. Masking helps optimize QR code reading by removing unwanted patterns that can interfere with reading.

Version information (optional): For larger QR codes, there are also version information cells that inform the reader about the size of the QR code.

Data and error correction codes: The internal area of QR codes stores data and error correction codes. Data is encoded in binary form, while error-correcting codes allow the data to be read even if the QR code is damaged or partially obscured.

Quiet Zone: This is an empty white frame that surrounds the QR code and helps the reader isolate the code from the surrounding content.

When reading QR codes, optical sensors detect black and white cells as binary data (1 for black cells and 0 for white cells). The data is then decoded and interpreted. QR codes can store text data, URLs, email addresses, phone numbers, and more. In addition, QR codes support Reed-Solomon error-correcting coding, which can recover data if the code is damaged or partially unreadable [3,4,17,18].

The proliferation of QR code reader apps on smartphones and mobile devices has further increased the use of QR codes in everyday life as people can easily access data using QR codes.

B. The basics of image processing and its relevance in the recognition of QR codes

Image processing is a set of techniques and algorithms used to analyze and modify images to extract desired information or improve image quality. Image processing is key to recognizing and decoding QR codes, since QR codes are most commonly found in the form of images and are scanned by optical devices such as cameras [19].

C. The process of image processing

Image input: The first step in the QR code recognition process is to scan the image, often performed by a camera or scanner. The scanned image is stored in digital form and represents the original QR code based on the intensity values of each pixel.

Preprocessing: Preprocessing of the scanned image may include noise reduction, sharpness enhancement, and contrast normalization. This helps improve the quality of the image and makes it easier to determine the position and data content of the QR code. Preprocessing is an important step in image processing that is usually performed before image analysis to improve image quality and reduce noise. During preprocessing, various techniques are used to enhance images and prepare them for analysis. The preprocessing steps and methods for QR codes can be as follows:

- **Grayscale conversion:** Convert color images into grayscale images. This reduces the amount of data in the image and makes analysis easier. It converts the intensity of each pixel of a color image into a single value that represents brightness.
- **Noise filtering:** Noise filtering is used to remove unwanted patterns in the image. Common noise filtering methods include the median filter, which takes the middle value of pixel values from the values of neighboring pixels, and the Gaussian filter,

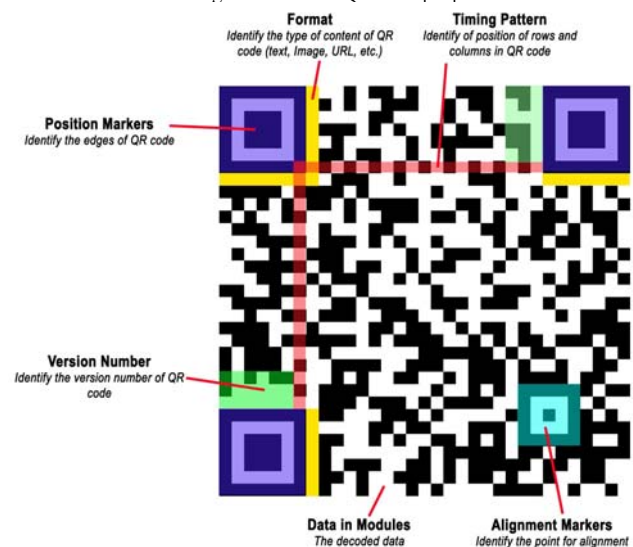
which weights pixel values by the values of neighboring pixels [21].

- **Contrast Manipulation:** Increase or decrease contrast to bring out image details. Adaptive histogram equalization is a commonly used method to increase contrast [22].
- **Edge detection:** Detect and highlight edges in the image to make it easier to identify the frame and position of the QR code. Canny edge detector and Sobel operator are common edge detection methods [23].
- **Binary Threshold Application:** Convert a grayscale image to a black and white image. This makes QR codes easier to identify and segment from the background [24].
- **Morphological operations:** Used to remove small noises and highlight features of QR codes. Morphological operations include dilation (widening) and erosion (narrowing) [25].
- **Perspective correction:** Correction of possible distortions that may arise from the angle or distance of the camera. This makes the QR code flat and easier to read [26].
- **Image segmentation:** The image is divided into different parts to distinguish the QR code from the background. Segmentation helps you highlight and analyze your code [27].
- **Improve Image Sharpness:** Sharpen the image to make the QR code easier to read. Improving sharpness often involves anti-aliasing and sharpening [28].
- **Data normalization and scaling:** Scaling and normalizing image data to optimize QR code recognition and decoding. Normalization usually involves rescaling pixel values to a specific range (eg 0-1) to provide a consistent data format [29].
- **Frame recognition:** QR codes have special patterned frames that help identify them in the image. Frame recognition algorithms identify these patterns and their position in the image [30].
- **Decoding:** After the QR code is cut and prepared, the information content of the image is transformed into meaningful data during the decoding process [31].

Numerous researches and literature support the use of listed image processing techniques in the implementation of QR code recognition [32,33].

The steps and methods mentioned above are not always necessary in every QR code preprocessing process, and the techniques used may vary depending on the specific task and the image quality available. The purpose of preprocessing is to optimize images for QR code recognition and decoding, and to reduce factors that make QR codes difficult to read, such as noise, blur, distortion, and low contrast.

Figure 1. Parts of QR code [14]



Position detection: QR codes have three large square markers located in the three corners of the code. Image processing helps identify these markers and thereby determine the position and orientation of the QR code in the image. For recognition of positioning patterns in QR codes, unique ratios are used, such as 1:1:3:1:1 (Figure 1.), which represents the width of the black and white areas in the pattern. Recognizing the position of QR codes is a key step in successfully reading and decoding the codes. The position recognition patterns are found in the upper left, upper right and lower left corners of the QR codes. These patterns help the reading algorithm determine the exact position and orientation of the QR code on the image, which is important because QR codes can be read from different angles and distances. The position recognition patterns are recognized by the reader in the following way:

- **Pattern recognition:** The reader looks for a special pattern in the image that is similar to the recognition pattern of the QR code position. This pattern uses a ratio of 1:1:3:1:1, which means that the pattern consists of black, white, black, white, and black again, with their widths proportional to each other.
- **Position determination:** Once the reader recognizes the position recognition patterns, it uses them to determine the orientation and position of the QR code in the image. This also allows the reader to eliminate image distortions and rotations and correctly orient the QR code for analysis.
- **Geometric transformation:** The reader can apply geometric transformations such as rotation and scaling to optimize the position of the QR code during reading and decoding.
- **Decoding:** After the reader has determined the position and orientation of the QR code, it analyzes and decodes the pixels of the code to extract meaningful information from it.

The use of position recognition patterns allows QR codes to be read quickly and efficiently in a variety of conditions,

even if the code is partially obscured, distorted or rotated. The distinctive ratio of the patterns, i.e. 1:1:3:1:1, allows the reader to uniquely identify the position recognition patterns without confusing them with other parts of the image or other patterns. This greatly contributes to the speed and reliability of reading QR codes.

Perspective correction: Perspective correction is key to successfully decoding QR codes, especially when the code has been photographed from an oblique angle. The main goal is to transform the image data in such a way that the QR code appears as a flat surface, as if we were directly facing it. Recognizing the check boxes helps identify the position and orientation of the code. These markers also help with perspective correction. In the case of perspective correction, the homography matrix can be used well. Homography is a geometric transformation that can represent perspective distortion. By recognizing the position of the markers, we can calculate a homography matrix that describes how the image changes due to the distortion. With the help of the homography matrix, we can transform the image back in such a way as to correct the perspective distortion. We do this by applying the inverse homography matrix to the pixels in the image and resampling it applying the reparametrization given by homography matrix.

Convert to binary image: After perspective correction, the image must be converted to binary form, which means it contains only black and white pixels.

- **Convert to grayscale image:** First, the image is converted to grayscale form, which means that each pixel has only one intensity value (not three different color components).
- **Determining a threshold value:** To create a binary image, a threshold value must be defined. Pixels with an intensity lower than this value will be black, while pixels with a higher intensity will be white. The Otsu method is an efficient way to automatically find the best threshold value.
- **Binarization:** After determining the threshold value, apply it to the image to obtain a binary image.

Decoding: This is followed by the decoding of the QR code, which involves reading the cells of the QR code and decoding the data stored by the QR code. The information hidden in QR codes is usually encoded with a Reed-Solomon coding scheme, which is also capable of error correction.

Finally, the data modules are identified and decoded from the QR code. During decoding, image processing algorithms identify the QR code pattern and convert it into data such as text or URLs.

When the QR code data has been successfully decoded, the application or device processes the data appropriately, such as opening the web page URL or displaying the text stored in the QR code.

The steps described above are general steps and different QR code readers and software may use different algorithms and methods to recognize and decode QR codes.

Error correction: QR codes often include error correction codes that allow the system to recover data if the code is corrupted or incomplete. Image processing also plays a role here, as it helps identify and correct any errors during data decoding.

Image processing techniques such as noise filtering, contrast manipulation, and edge detection are essential in preprocessing QR codes so that they can be properly decoded. Also Reed-Solomon codes provide error correction, allowing QR codes to be reliably readable even in damaged or noisy environments.

Image processing is essential for efficient recognition and decoding of QR codes. Preprocessing of images scanned by cameras and scanners, identification of the position and orientation of the QR code, perspective correction, binary conversion and decoding are all part of image processing. All of this allows QR codes to be scanned quickly and efficiently and data to be made available to users.

IV. CONCLUSION AND DISCUSSION

Image processing plays an important role in recognizing and decoding QR codes. This process includes scanning the images in digital form, preprocessing them with noise reduction, contrast enhancement, and edge detection, as well as determining the position and content of the QR code. Image processing steps include binarization, morphological operations, perspective correction, and image segmentation, all of which contribute to the efficient recognition and reading of QR codes. Position recognition, application of geometric transformations, and the decoding process are key elements of image processing that enable quick and accurate interpretation of QR codes. Image processing is thus a fundamental technology in working with QR codes, helping to access information easily and efficiently.

Modern image recognition is done using convolutional neural networks (CNN). Modern image recognition technologies such as Convolutional Neural Networks (CNN) significantly improve the ability to recognize QR codes. Convolutional neural networks are deep learning models that are particularly well suited to image recognition tasks. CNNs can learn and recognize patterns, textures, and objects in images, including QR codes.

CNNs are able to "learn" about the characteristics of QR codes through large data sets. This means that the network can learn and recognize the patterns and structures characteristic of QR codes, such as the square shape and coding patterns. This capability enables the network to efficiently and accurately recognize QR codes even when the image quality is not ideal or when the code is partially obscured [20].

Deep learning can adapt to different environmental conditions, such as lighting and image quality. CNNs are able to "learn" the effects of different lighting conditions and image quality and adapt to these changes. This means that the network may be able to recognize QR codes even if the image is too light or too dark, or if the image quality is low.

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